

Exercise 1 - SQL Fundamentals

SQL Fundamentals: Set SELECT & Filtering

Section 1: Selecting & Sorting

Question 1

SELECT *

FROM employees;

employees

id	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Question 2

SELECT DISTINCT department

FROM employees;

employees

department
IT
HR
Finance
Marketing

Question 3

```
SELECT first-name,  
       last-name  
FROM employees  
ORDER BY salary DESC;
```

employees

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Question 4

```
SELECT id,  
       first-name,  
       last-name,  
       salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

employees

id	first-name	last-name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Section 2: Filtering with WHERE

Question 4.5

```
SELECT id,  
       first-name,  
       last-name,  
       department
```

```
FROM employees
```

```
WHERE department = 'IT';
```

employees

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Question 6

```
SELECT id,  
       first-name,  
       last-name,  
       department,  
       salary
```

```
FROM employees
```

```
WHERE department = 'Finance' AND salary > 60000 60000
```

employees

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

Question 7

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department = 'HR'  
WHERE department IN ('HR', 'Marketing');  
employees;
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Question 8

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department NOT IN ('IT');  
employees
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Question 9

```
SELECT id,
       first-name,
       last-name,
       department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first-name	last-name	department
2	Brian	Lee	
3	Cathy	Zulu	
4	David	Mokoena	Marketing

employees.

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Question 10.

```
SELECT id,
       first-name,
       last-name,
       department,
       salary,
       city
```

FROM employees

WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg